The Use of Context-Sensitive Insurance Telematics Data in Auto Insurance Rate Making

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Abstract

Historically, auto insurers use various socio-demographic underwriting factors to differentiate driver risks. With the invention of GPS devices, information such as mileage, traffic condition and driving habits can be incorporated into auto insurance premium calculation. Several major auto insurance companies have offered usage based insurance (UBI) programs where auto insurance premiums are sensitive to actual GPS readings combined with driver's driving behavior. However, given that telematics data are proprietary to the insurance companies that collect such data, the accessibility of UBI is extremely limited.

In this research we showcase how second-by-second GPS data can be integrated into existing or new auto insurance pricing structures. We use two types of data: real-time GPS trajectory data collected using a traffic app, as well as survey data. We incorporate vehicle trajectories and accident data to quantify the relationship between driving hazard and accidents with the goal of establishing the linkage between driving risks and accident costs. GPS data considered in this study include not only tradition UBI factors, but also the unique contextual-based risk measurements that compares the driving speed of the vehicle with other drivers on the same road segment. Although smartphone data is used in this study, the methodology developed can be applied to GPS trajectory data collected from other devices such as on-board diagnostics (OBD) or black box solutions. We find that hard brakes, hard starts, peak time travel, speeding as well as driving at a speed significantly different from traffic flow are highly correlated with accident rate. We illustrate the potential underwriting loss an insurer may incur resulting from adverse selection if omitting pertinent risk factors. The results of our study can help insurance companies that are interested in getting into the UBI area set up their auto insurance premium rates.

Keywords: Insurance rate making; usage based insurance; insurance telematics; smartphone data; context-sensitive data

1. Introduction

The auto insurance industry has been changing rapidly in recent years due to advances in telematics technology. Historically, auto insurers use various socio-demographic underwriting factors to differentiate driver risks. Even though studies have shown that mileage, traffic congestion and facility attributes are also important attributes to auto accidents, they have not been used in premium rating simply because data has not been available. With the invention of GPS devices, such information becomes accessible. In the recent years, several major auto insurance companies have offered usage based insurance (UBI) programs where auto insurance premiums are sensitive to actual GPS readings combined with a driver's driving behavior. The Brookings Institution suggests that UBI insurance programs incentivize people to cut down unnecessary driving which could potentially yield $50-$60 billion in terms of social benefits. However, insurance companies need to overcome several challenges before they can fully adopt an UBI program. In addition to the high costs associated with the technology, one major challenge insurers are faced with is the unfamiliarity with the insurance telematics data.

In this research we showcase how GPS data can be integrated into existing or new auto insurance pricing structures. We use two types of data: real-time GPS trajectory data collected using a unique traffic app, as well as survey data on driver background and loss history. Note that although smartphone data is used in this study, the methodology developed can be applied to GPS trajectory data collected from other devices such as on-board diagnostics (OBD) or black box solutions. One uniqueness of our study is that we incorporate vehicle trajectories, associated link traffic volume and speed data, time of day information, and accident data to quantify the relationship between hazard, driving behavior, and auto accident with the goal of establishing the linkage between driving risks and accident costs. Our analysis is based on a total of 131,537 trips taken by 503 drivers over an 18-month period. We first examine the characteristics of our second-by-second trajectory data and determine the appropriate buckets for creating risk variables based on each variable's univariate distributions. Next, we create trip level measures and use different thresholds for defining risk levels. In our statistical models we incorporate static factors used in traditional auto insurance pricing such as gender, age, and vehicle mileage, the common UBI risk factors such as time of travel, hard brake, and hard start, and also the unique contextual-based risk measurements that are proposed and studied for the first time; namely speeding and relative speed over various context data such as link speed, congestion level, and speed limit. We model both the probability of accidents as well as number of accidents, calculate the probability of accidents considering corresponding risk factors, analyze the relative probability ratios and incidence-rate ratios for drivers in each risk category, and compute and compare actuarially fair premiums for drivers corresponding to their risks. We also illustrate the potential underwriting loss resulting from adverse selection, if an insurance company does not price auto insurance according to a driver's respective driving risk factors. As expected, the probability of auto accident is lower for female, older drivers and those that have more driving experience. We also find that vehicle mileage travelled, hard brake, hard start, peak time travel, speeding as well as driving at a speed significantly different from traffic flow are highly correlated with accident rate. According to our analysis, insurers that omit these individualized driving factors could undercharge their policyholders up to 30.1%. The results of our study can help insurance companies that are interested in getting into the UBI area set up their auto insurance premium rates. The findings revealed by our analysis also establish the actuarial foundation for auto insurers seeking approval for the use of UBI rating plans from their respective state authorities.

The paper is organized as follows. The next section reviews related literature on adverse selection and UBI programs, followed by research methodology that shows the data sources, variable creation and empirical models. The next section is the empirical results that show the generalized linear models (GLM) models and prediction of insurance premiums. We conclude with the main findings as well as the implications and limitations.

2. **Literature review**

Moral hazard and adverse selection have long been the major concerns for the insurance industry. Within the insurance context, moral hazard refers to a situation where policyholders have lower motivation to prevent loss and/or reduce loss severity in the presence of insurance (see, for example, Arrow 1963; Pauly 1968; and Zeckhauser 1970). Once a person has acquired auto insurance, there is little financial incentive to drive less or more cautiously, unless premiums are reflective of such behavior. To control moral hazard problems, insurance companies typically build in provisions in insurance contracts and factor in additional premiums attributable to moral hazard if the behavior of their policyholders can be identified and measured. The challenge here is the identification of policyholder behavior. Typical underwriting factors for auto insurance include driver characteristics such as age, gender, prior driving experience and information of vehicle, etc., as data on those factors are easily available. Driving habits, which are often key attributes to accidents, have not been incorporated into actuarial pricing until recently simply because such data is not available and/or the cost of obtaining such information is too high.

When insurers are not able to differentiate high risks from low risks by charging different premiums, adverse selection becomes a concern. Rothschild and Stiglitz (1976) pioneered a theoretical model on adverse selection describing market equilibrium when asymmetric information exists in the insurance market. They argue that asymmetric information leads to either a separating equilibrium, where high-risks and low-risks purchase policies with varying amounts of coverage, or to no equilibrium. If the market does achieve an equilibrium, the high-risks purchase full coverage while the low-risks purchase partial coverage. Wilson (1977) and Miyazaki (1977) also describe market equilibria under different assumptions from the Rothschild and Stiglitz model. Following the theoretical prediction, several studies examined the correlation between risk levels and insurance coverage (see, for example, Puelz and Snow 1994; Dionne et al. 2001; Cohen 2005; Saito 2006 et al. 2010).

With advances in technology, insurance telematics has become a hot topic in recent years. Several research papers have investigated the feasibility, costs and benefits of implementing UBI where auto insurance premiums are based on mileage travelled. Bordoff and Noel (2008) evaluated pay-as-you-drive auto insurance from the perspectives of user and social benefit, and propose a three-part strategy to overcome the barriers for wider implementation. Greenberg (2009) explore usage-based pricing and suggested a risk-adjusted mileage pricing to incorporate geography and driving habits with mileage driven, as opposed to a straight per-mile or per-minute pricing. Baecke and Bocca (2017) prove the value of telematics-based data in the risk selection process of an insurance company, where three models including a logistic regression, random forests and artificial neural networks model are built and compared. The simultaneous effect of the distance traveled and exposure time on the risk of accident is studied in (Boucher et al. 2017) by using Generalized Additive Models (GAM). Wahlström, et al. (Wahlström et al. 2015) detect dangerous cornering with global navigation satellite system (GNSS) measurements. More studies on insurance telematics can be found in Händel et al. (2014), Händel et al. (2015), Ohlsson et al. (2015), Girod et al. (2016) and Ferreira et al. (2017).

Ayuso et al. (2016) demonstrated how automobile insurance can be improved by incorporating mileage and driver behavior data. The key idea is that telemetry should facilitate the inclusion within insurance pricing of those factors that traffic authorities identify as being associated with risky drivers. The implication of risk-adjusted mileage pricing is also suggested by Ferreira and Minike (2010) using data collected from Massachusetts. Greenberg (2009) proposes and describes a regulatory incentive for insurance companies. Desyllasa and Sako (2013) examine how an incumbent firm profits from business model innovation through the study of Pay-As-You-Drive auto insurance. Fan and Wang (2017) compared underwriting decision making between telematics-enabled UBI and traditional auto insurance. Fernandes and Feero built a system to determine an initial insurance policy benefit based on smartphone data (2015).

Implementing a risk-adjusted insurance pricing could present some challenging issues. Mataija and Schoubroeck (2016) assess the legal challenges arising from the use of telematics technology in the insurance sector in the EU insurance market, in which legal challenges posed by telematics insurance, data protection and privacy, and other challenges facing telematics insurance including data ownership and indirect discrimination are identified. Litman (2012) compares several distance-based insurance pricing options and evaluates concerns and criticisms. The plans evaluated include mileage rate factor (MRF), pay-at-the-pump (PATP), usage-based premiums (per-mile or per-minute), and GPS-based pricing. The pricing alternatives are evaluated based on public policy concerns, such as actuarial accuracy, equity, affordability, road safety, consumer impacts, public acceptability, travel reduction impacts, congestion and facility cost savings, environmental protection, and economic efficiency and development. The analysis indicates that distance-based pricing is technically and economically feasible, and can provide significant benefits to motorists and society.

Currently, various types of UBI programs are provided by insurance companies in over 40 U.S. states and in other countries such as Australia, United Kingdom, Netherlands, South Africa, Israel, Spain, Japan, and Italy. For example, Progressive Insurance currently offers a program where policyholders may receive discounts in their auto insurance premiums if they agree to use a small device called the "Snapshot" that monitors their driving for a period of time. Other insurers such as State Farm, Allstate, Travelers, etc. all offer choices to their customers where insurance premiums are based on usage and/or driving behavior. It should be noted that different UBI solution providers have chosen different measurements to evaluate driver's performance. For example, Progressive uses a combination of hard braking, amount of miles driven, time and day, fast starts and trip regularity to calculate one's driving risk. They define a hard brake as deceleration over 7mph/sec (Progressive 2017).[1] The Hartford cites speed and braking as two of the factors that they consider when calculating premiums under their TrueLane program. Travelers differentiates premiums basing on driving speed, braking, acceleration, and time of day. Allstate's Drivewise program rewards drivers that limit high speed driving, late-night trips and hard braking. They consider driving speed higher than 80mph as unsafe. The offering of UBI thus far has been limited to large insurers in the marketplace mainly due to high cost and data availability. Recognizing the limitations this research establishes a framework for incorporating GPS data in the process of pricing auto insurance to help establish the UBI premium rates actuarially.

3. Data Description

In this research we use a smartphone based app called Metropia[2] to monitor a driver's driving behavior and collect GPS data. The multi-source data collected through the platform includes second-by-second GPS trajectory data of time, locations and speed, as well as time-varying traffic and roadway dynamic. One unique feature of the data is that the backend server estimates traffic speed and volume for each link that the vehicle traverses. Such data combining both user trajectory and link speed/volume information has rarely been seen in prior research, permitting a unique opportunity to link critical traffic factors that lead to certain driving behaviors and crash potential. The collected data allow us to not only identify how a driver drives, but also evaluate the performance in conjunction with information on the surrounding environment. In addition to real-time driving data, we also conduct a survey on active app users and collect information on the driver's demographic information as well as historical experience of auto accidents. Data collected via the survey include age, gender, driving experience, annual miles travelled, and whether the driver has been involved in an accident in the past three years. Combined with GPS driving performance data obtained, we analyze a driver's driving risks and provide a link between driving

[1] An anecdotal experiment in 2014 playing with Progressive snapshot smartphone app found that 75mph was used as a threshold to capture a speeding event, no matter where the vehicle is travelling.

[2] http://www.metropia.com/

risks and accident rate with the goal of establishing the actuarial foundation for auto insurance premium rating.

The GPS data collected include detailed vehicle dynamic information. When a user starts a trip, the internal GPS module is activated and starts to record the second-by-second latitude/longitude data location and instantaneous moving speed. These data allow detailed position, velocity, acceleration and deceleration details to be analyzed. The accuracy of GPS points can be affected by many factors including the quality of GPS module in the smartphone, network connections, tall buildings nearby, driving in the tunnels, etc. Generally speaking, when a vehicle is travelling on a road in an open environment, the GPS accuracy is typically 17ft (or 5 meters) which is reasonably good. We have followed the quality control procedure as described in Noh et al. (2016) to ensure that only good-quality data is included in data analysis. In a nutshell, map matching is performed to map GPS data points to road segment, and during this process low-quality data points are filtered out. After that, non-driving data are detected via a mode detection module, and off-street activities (e.g. coffee) are identified and filtered out through an activity filtering module. In terms of data collection frequency, the one-second sampling frequency is on par with OBD-II based solution implemented in the auto insurance industry (see, for example, Gagliordi 2014).

Table 1 shows the sample second-by-second trajectory data. The sample shows the data points collected within 6 seconds, from 7:58:11 AM to 7:58:16 AM, on April 9, 2016 on user #156. The recorded data includes the latitude, longitude and altitude of each particular GPS point, the heading of the vehicle as well as the instantaneous travel speed of the vehicle. Vehicle acceleration and deceleration speed at any given time can be calculated from the instantaneous moving speeds data. Additionally, the system stores dynamic traffic condition information, including instantaneous average speed of other vehicles and traffic volume of the current road segment. Each road segment has a unit link ID and speed limit for each link ID is also identified from the roadway geometry database.

Table 1 Sample GPS data

GPS ID	GPS Information							Network Information			
	User ID	Time Stamp	Latitude	Longitude	Altitude	Heading	Speed (mph)	Link ID	Link Speed (mph)	Link Volume (veh/hr)	Speed limit (mph)
1	156	4/9/2016 7:58:11 AM	32.250	-110.928	2351.0	272.1	34.7	9171	36.6	940	40
2	156	4/9/2016 7:58:12 AM	32.250	-110.928	2350.4	269.9	36.4	9171	36.6	940	40
3	156	4/9/2016 7:58:13 AM	32.250	-110.929	2349.4	269.8	36.9	9171	36.6	940	40
4	156	4/9/2016 7:58:14 AM	32.250	-110.929	2348.1	269.7	38.0	9171	36.6	940	40
5	156	4/9/2016 7:58:15 AM	32.250	-110.929	2346.1	269.9	38.6	9171	36.6	940	40
6	156	4/9/2016 7:58:16 AM	32.250	-110.929	2345.5	269.9	38.6	4584	37.1	890	35

It should be acknowledged that different data collection methods come with pros and cons. Handel et al. (2014) provides a summary of how smartphone-based UBI solutions compare with the classical OBD or black box solutions. For example, the speed attribute from smartphone could be more accurate than OBD-based solutions, whereas OBD-based solution is superior in terms of integrality, availability and continuity of services. The merits and limitations of different data collection methods have also been analyzed from a total of 12 aspects including acceleration, braking, speeding, etc. Although we use smartphones to collect data, the methodologies developed in this study are applicable to GPS trajectory data collected using other devices such as on-board diagnostics (OBD) or black box solutions.

4. Research Methodology

For convenient reference purposes, the mathematical notations used in this section are presented below first.

t: timestamp

l: link index

i: trip index

V_t: user's driving speed at time t

$\bar{v}_{t,l}$: average driving speed on link l at time t

HB and HS: hard brake and hard start

$Speeding$: speeding event

r_t: relative speed value which takes two forms of r_t^a and r_t^t

r_t^a: relative speed in absolute value

r_t^t: relative speed in percentage

Rel_Spd_t: relative speed flag

R_t: contextual relative speed

P_t : a general term that represents trajectory level measures discussed above

P^i: a general term that represents trip level measures for trip i calculated by weighted average of P_t

$threshold_{HB}$, $threshold_{HS}$, $threshold_{rel}$, $threshold_{speeding}$: thresholds to determine hard brake, hard start, relative speed and speeding, respectively

$dist_t$: distance traveled between timestamp $t+1$and t

acc_t: instantaneous change of speed at timestamp t

$FFS_{t,l}$: free flow speed of link l and timestamp t

$c_{t,l}$ is congestion level at timestamp t on link l at time t

s_t: value of speed over speed limit

$spdlimit_{t,l}$: posted speed limit for link index l the vehicle is located at timestamp t.

F: a defined data fusion function to integrate r_t and $\bar{v}_{t,l}$ to contextual relative speed measure R_t

G: a defined data fusion function to integrate s_t and $c_{t,l}$ to contextual speeding performance measure S_t

$\delta_{m,i}$: expected probability of an accident for the mth driver on the ith trip, conditional on other factors

π_m : an array of driver information that are trip invariant such as age, gender, and driving experience

$\gamma_{m,i}$: a matrix of standardized trip-level variables created from GPS data.

a and λ : two coefficients used in GLM function

$\delta(p)$: loss odds ratio

$\varepsilon_{m,i:}$ a term representing random error

p : probability of having a loss

4.1. Performance measurements

We create four measurements from GPS data to capture driving performance, including hard brake, hard start, speeding and relative speed.

Hard brake and **hard start** measures the acceleration and deceleration process in the vehicle operation. Hard brake flag

$$HB_t = 1, \text{ if } acc_t < threshold_{HB} < 0 \quad \text{(Eq. 1)}$$

and hard start flag

$$HS_t = 1, \text{ if } acc_t > threshold_{HS} > 0 \quad \text{(Eq. 2)}$$

where

$$acc_t = V_{t+1} - V_t \quad \text{(Eq. 3)}$$

Different thresholds can be used to define qualified hard brakes and hard starts. The distribution of acceleration and deceleration from random sampling out of the 162 million GPS points collected is displayed in Figure 1.

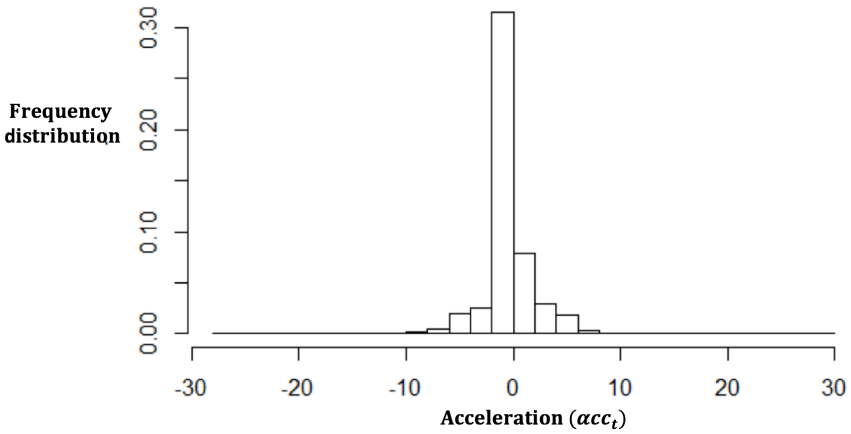


Figure 1 Distribution of acceleration acc_t (ft/s^2)

The X-axis represents the instantaneous change of speed, with positive values indicating acceleration and negative values indicating deceleration. The Y-axis is the percentage of GPS point data that fall into each bin. As can be seen, the number of observations with deceleration <-10ft/s^2 or acceleration >10ft/s^2 is very limited. Only less than 1% of the observations have extreme values for acceleration or deceleration.[3] More details on hard brakes and hard starts will be discussed in section 4.2.

Relative speed measures the variations between a driver's driving speed and the real-time traffic speed on the same road segment. Variation can be measured by absolute deviation as well as relative deviation. For absolute deviation, we calculate the following:

$$r_t^a = V_t - \bar{v}_{t,l} \quad \text{(Eq. 4)}$$

Relative speed can also be measured in terms of percentage deviation from link speed. For this form of measurement, we calculate the following:

[3] The percentage of observations with deceleration <-10ft/s^2 is 0.14% and the percentage of observations with acceleration >10ft/s^2 is 0.12%.

$$r_t^t = \frac{V_t - \bar{v}_{t,l}}{\bar{v}_{t,l}} * 100 \quad \text{(Eq. 5)}$$

A higher relative speed value stands for a larger discrepancy in the driving behaviors between this driver and other drivers on the same link, i.e. this driver either drives much faster, or much slower compared with other drivers, which indicates unsafe driving behavior. Flags are created when r_t (which takes the forms of r_t^a and r_t^r, respectively) is greater than specified thresholds. Specifically,

$$Rel_Spd_t = 1 \text{ if } r_t > threshold_{rel} \quad \text{(Eq. 6)}$$

where $threshold_{rel}$ can be expressed in term of miles per hour, or percentage deviation.

Figure 2 shows the distribution of relative speed. In this figure, the X-axis represents the relative speed, with positive values indicating users driving faster than traffic flow and negative values indicating users driving slower than traffic flow.

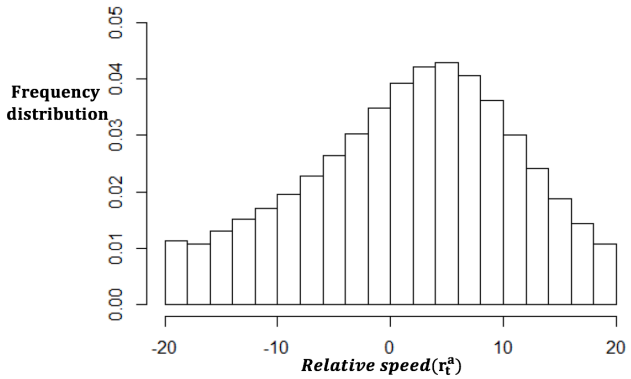


Figure 2 Distribution of relative speed r_t^a

In general, we can tell the curve skews slightly to the left, which indicates driving faster than the traffic speed is more commonly observed. Driving at a speed that is about 4~6mph higher than the traffic speed on the road is the most frequently observed behavior in our sample.

Considering the contextual concept in the driving behavior, the same variation r_t may have different risky impact at slow and fast traffic flow conditions. For example, driving at a speed that is 10mph faster than others where the average speed is 65mph may be more dangerous than when the average traffic speed is 35mph. Therefore, the contextual relative speed R_t is constructed as

$$R_t = F(r_t, \bar{v}_{t,l}) \quad \text{(Eq. 7)}$$

where the relative speed is a function of speed deviation from traffic speed at link *l,* as well as the traffic speed itself.

Speeding captures the tendency of driving at a speed significantly faster than the posted speed limit. We first calculate the difference between the vehicle's driving speed and speed limit, and create flags if the difference is greater than the specified threshold. Specifically,

$$Speeding_t = 1 \text{ if } s_t > threshold_{speeding} \quad \text{(Eq. 8)}$$

where

$$s_t = \max(V_t - spdlimit_{t,l}, 0) \quad \text{(Eq. 9)}$$

Similar to hard brakes and hard starts, different thresholds can be specified for identifying speeding.

Figure 3 shows the distribution of speeding (i.e. driving faster than speed limit). In this figure, only the speeding samples are visualized, with the X-axis representing the degree of speeding and the Y-axis indicating the percentage of samples in the corresponding bin. As can be seen, most of speeding are within 5 miles over speed limit, and there are only a few cases where speeding is more than 15 miles over speed limit.

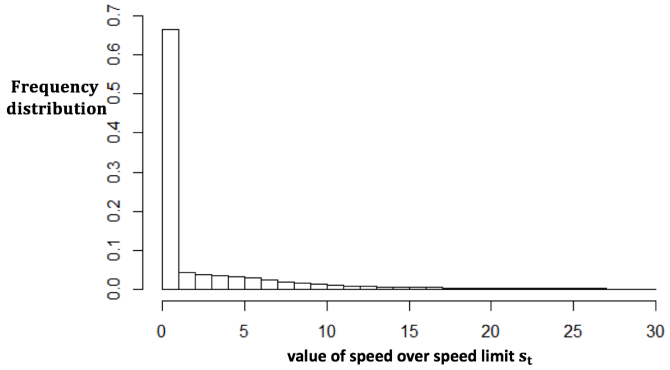


Figure 3 Distribution of value of speed over speed limit s_t

To reflect true risk, speeding should be adjusted with contextual status, such as speed limit and/or congestion levels. For example, 10mph over speed limit may impose lower risks on freeway than on arterials. Similarly, the risk imposed by speeding may vary between congestion and free flow conditions. A contextual speeding performance measure that captures the both the speeding and traffic conditions can be written as follows:

$$S_t = G(s_t, c_{t,l}) \quad \text{(Eq. 10)}$$

where

$$c_{t,l} = \bar{v}_{t,l} / FFS_{t,l} \quad \text{(Eq. 11)}$$

and $c_{t,l} \in [0,1]$. $c_{t,l} = 1$ represents free flow condition and $c_{t,l} = 0$ if completely congested.

4.2. **Trip level measurements**

Using GPS data, various trajectory risk measurements can be calculated at the trip level. Each of the above trajectory level measures is aggregated for each trip to create trip level data. Trip level performance can be considered as weighted average with travel distance $dist_t$ as the weight value. Specifically,

$$P^i = \frac{\sum_t (dist_t * P_t)}{\sum_t dist_t} \quad \text{(Eq. 12)}$$

All trip level measures are standardized by distance travelled to ensure comparability of data; for example, the number of hard brakes per mile, or the percentage of a trip when a driver is speeding, etc.

As industry practice on definitions for hard acceleration and deceleration have not been established, we used multiple thresholds to identify risky behavior. Given the observed speed distributions, we choose three different cutoff points for defining hard brakes and hard starts: deceleration <-15ft/s^2, <-12ft/s^2, or <-10ft/s^2 for hard brakes, and acceleration >15ft/s^2, >12ft/s^2, or >10ft/s^2 for hard starts. In this section, we show how distributions change by choice of threshold. Using different thresholds to define risky behavior allows insurers to consider alternative risk classification systems. In section 5, we illustrate how insurance premiums should be adjusted corresponding to individualized risk profiles.

Table 2 shows the distributions of the variables hard brake and hard start using different thresholds. Both hard brakes and hard starts are rare as most trips do not have qualified hard brakes or hard starts. When we use -10ft/s^2 as the threshold, the average number of qualified hard brakes is 0.132 per mile, which is equivalent to 13 hard brakes per 100 miles. The average number drops to 0.064 and 0.02 when we increase the threshold to -12ft/s^{2}and -15ft/s^2, respectively. At the 75th percentile, the number of qualified hard brakes are 0.193, 0.078, and 0 when we use -10ft/s^2, -12ft/s^2 and -15ft/s^2 as the threshold, respectively. Hard starts follow a similar pattern.

Table 2 Distribution of hard brake and hard start

	Hard Brake			Hard Start		
Thresholds	-10 ft/s^2	-12 ft/s^2	-15 ft/s^2	10 ft/s^2	12 ft/s^2	15 ft/s^2
Mean	0.132	0.064	0.020	0.124	0.067	0.023
100% Max	5.482	2.400	1.677	4.127	3.090	2.060
99%	1.043	0.647	0.326	0.976	0.665	0.360
95%	0.573	0.332	0.151	0.533	0.341	0.171
90%	0.391	0.225	0.073	0.369	0.230	0.087
75%	0.193	0.078	0	0.181	0.085	0
50%	0	0	0	0	0	0
5%	0	0	0	0	0	0
10%	0	0	0	0	0	0
5%	0	0	0	0	0	0
1%	0	0	0	0	0	0
0% Min	0	0	0	0	0	0

At the trip level, the relative speed variable reflects the percentage of a trip where a user's driving speed significantly deviates from the link travel speed. Driving faster or slower than flow of traffic may be viewed as a hazard. We consider not only the direction of relative speed (e.g. faster or slower), but also the differentiation effect at different levels of link speed. Two types of thresholds are considered: *absolute deviation* and *relative deviation*. Table 3 and Table 4 show the distribution of relative speed. Relative speed is defined using absolute mileage (e.g. 10mph deviating from link speed) in Table 3, and using percentage deviation (e.g. 30% deviation from link speed) in Table 4.[4]

Table 3 Distribution of relative speed when driving speed deviates from link speed by 10mph

	All	Faster than link speed			Slower than link speed		
		Both	Link < 40mph	Link ≥ 40mph	Both	Link < 40mph	Link ≥ 40mph
Mean	0.382	0.279	0.218	0.061	0.103	0.074	0.029
Max	1.000	1.000	1.000	1.000	1.000	1.000	0.968
99%	0.806	0.750	0.688	0.494	0.437	0.302	0.268
95%	0.678	0.600	0.532	0.288	0.270	0.176	0.147

[4] We have tried several other alternatives. For absolute deviation, we also use 5mph deviation from link speed. For relative deviation, we use both percentage deviation and standard deviation. For percentage deviation, we use 30%, 50%, 70% and 100% deviation from link speed. For standard deviation, we choose 1 and 2 standard deviations from link speed. The results from these alternative deviations can be obtained from the authors.

90%	0.607	0.519	0.449	0.195	0.203	0.135	0.097
Median	0.494	0.389	0.316	0.080	0.123	0.091	0.035
25%	0.257	0.147	0.089	0	0.052	0.039	0
10%	0.163	0.065	0.029	0	0.035	0.022	0
5%	0.119	0.026	0.007	0	0.026	0.014	0
1%	0.066	0	0	0	0.012	0.005	0
Min	0	0	0	0	0	0	0

Table 4 Distribution of relative speed when driving speed deviates from link speed by 30% of link speed

	All	Faster than link speed			Slower than link speed		
		Both	Link < 40mph	Link ≥ 40mph	Both	Link < 40mph	Link ≥ 40mph
Mean	0.351	0.276	0.247	0.030	0.075	0.072	0.003
Max	1.000	1.000	1.000	1.000	1.000	1.000	0.653
99%	0.809	0.751	0.733	0.323	0.310	0.293	0.059
95%	0.677	0.606	0.585	0.144	0.182	0.171	0.020
90%	0.603	0.525	0.503	0.088	0.137	0.131	0.009
75%	0.474	0.390	0.361	0.033	0.091	0.088	0
Median	0.334	0.252	0.216	0	0.061	0.059	0
25%	0.209	0.139	0.104	0	0.040	0.038	0
10%	0.122	0.060	0.035	0	0.023	0.021	0
5%	0.086	0.024	0	0	0.015	0.014	0
1%	0.046	0.000	0	0	0.006	0.005	0
Min	0	0	0	0	0	0	0

Percentage of a trip deviating from link speed by more than 10mph ranges from 0 to 100%, meaning some drivers constantly drive at a speed either significantly faster than link speed or significantly slower than link speed. On average, 38.2% of a given trip will have a speed either faster or slower than link speed by at least 10mph. When looking at directional deviation, on average, 27.9% of a trip have a speed faster than link speed by 10mph, and 10.3% of a trip have a speed slower than link speed by 10mph. When separating by link speed, there are more deviations from link speed when link speed is slower than 40mph in comparison to situations when link speed is 40mph or faster. For example, a driver's average driving speed is faster than link speed on 21.8% of a trip when link speed is slower than 40mph. However, the average drops to 6.1% when link speed is 40mph or faster.

Similarly, the percentage of a trip where a driver's driving speed is slower than link speed is a lot lower when link speed is 40mph or higher than when link speed is lower than 40mph. These results suggest that speed variation by more than 10mph is less likely when link speed is faster. When we use 30% deviation from link speed to define driving speed significantly deviating from link speed, the results show a similar pattern with some variations. On average, 27.6% of a given trip has a driving speed faster than link speed, where on average 24.7% takes place when link speed is slower than 40mph, and only 3% takes place when link speed is faster than 40mph. Similarly, 7.5% of the trip has a driving speed slower than link speed, and almost all of the cases are from trips where the link speed is slower than 40mph. In other words, it is very unlikely to have driving speed that is 30% slower than link speed when link speed is 40mph or higher.

For the speeding factor, we consider not only the driving speed, but also the speed limit as well as congestion level. We consider three levels of speeding using 10mph over speed limit, 8mph over speed

limit, and 5mph over speed limit. Table 5 and Table 6 show the distribution of the speeding variable using 10mph and 8mph as the thresholds.[5] The speeding variable reflects the percentage of a trip where the driving speed exceeds the speed limit by the specified amount. For example, on average, 2.9% of a trip shows a speed over the speed limit by at least 10mph, and 4.8% of a trip shows a speed over the speed limit by at least 8mph. As expected, it is more likely to exceed the speed limit by 8mph than by 10mph, evident by greater figures for all cells associated with 8mph than for cells associated with 10mph.

When separating by speed limit, we can see that speeding is more likely when the speed limit is below 60. For example, when the speed limit is 60 or above, only 0.3% of a trip has a driving speed 10mph beyond the speed limit. The percentage increases to 2.5% when the speed limit is below 60. Similarly, speeding is much more likely when there is no congestion. During congestion, on average, only 0.3% of a trip has a driving speed that is 10mph beyond the speed limit. This probability increases to 2.6% when there is no congestion. During traffic congestion, a maximum of 76.9% of a trip where the driving speed is beyond the speed limit by at least 10mph is observed. When there is no congestion, the same percentage can go as high as 100% of a trip. In other words, someone could have a driving speed that is constantly beyond the speed limit for the whole trip.

Table 5 Distribution of speeding: speeding defined as 10mph over speed limit

	All speeding	By speed limit		By congestion	
		Speed limit < 60mph	Speed limit ≥ 60mph	Congestion	No Congestion
Mean	0.029	0.025	0.003	0.003	0.026
Max	1.000	1.000	0.714	0.769	1.000
99%	0.278	0.244	0.093	0.051	0.268
95%	0.128	0.110	0.013	0.021	0.117
90%	0.082	0.071	0	0.010	0.072
75%	0.033	0.030	0	0	0.028
Median	0.004	0.004	0	0	0.003
25%	0	0	0	0	0
10%	0	0	0	0	0
5%	0	0	0	0	0
1%	0	0	0	0	0
Min	0	0	0	0	0

Table 6 Distribution of speeding: speeding defined as 8mph over speed limit

	All speeding	By speed limit		By congestion	
		Speed limit < 60mph	Speed limit ≥ 60mph	Congestion	No Congestion
Mean	0.048	0.041	0.006	0.005	0.043
Max	1.000	1.000	0.791	0.773	1.000
99%	0.371	0.330	0.149	0.068	0.355
95%	0.192	0.163	0.039	0.028	0.178

[5] To save space, we only report the distributions for 10mph and 8mph. The distribution using 5mph as the threshold follows the same pattern and is available up request from the authors.

90%	0.131	0.110	0.006	0.015	0.119
75%	0.062	0.053	0	0	0.054
Median	0.018	0.016	0	0	0.014
25%	0	0	0	0	0
10%	0	0	0	0	0
5%	0	0	0	0	0
1%	0	0	0	0	0
Min	0	0	0	0	0

In addition to the above four measures, GPS data also reveal the time when each trip takes place. Figure 4 shows the distribution of when each trip takes place. We create a trip level variable identifying whether the trip takes place during peak hours (i.e. 7:00-9:00am or 5:00-7:00pm) or not. For trips that span across peak hours and non-peak hours, we use 50% as the threshold. If more than 50% of one trip takes place during peak hours, we consider the trip to be taken during peak hours.

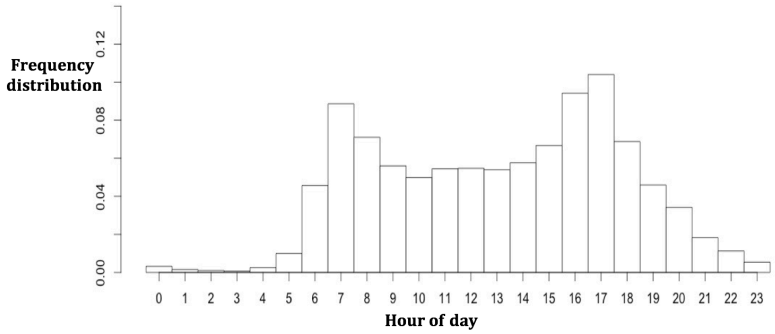


Figure 4 Distribution of time of travel

4.3. Sample Selection

The selection of participants for this research is based on an opt-in scheme, where all frequent users, whoever completed more than 10 trips with the App, are invited to report demographic information, driving experience and accident history. Our sample begins with a total of 525 drivers. Collectively, these drivers completed 136,244 trips during our sample period from January 2015 to July 2016. There are only 7 drivers younger than age 20 and 15 drivers older than age 60, and all of them did not have any accident in the past year. Given the small number of drivers in these age categories, combined with zero (or close to zero) accidents incurred by these drivers, we have removed them from our sample so our results are not skewed by these observations. As a result, our final sample includes a total of 503 drivers. Table 7 shows the driver characteristics of our sample.

Table 7 Basic statistics of driver characteristics

Variable	Number of Drivers	Percentage of Drivers
Number of at fault accidents		
0	431	85.7%
1	65	12.9%
2	7	1.4%
Driving experience 10+ years	419	83.3%
Female	264	52.5%
Age		

< 30	145	28.8%
30 to 44	231	45.9%
45 to 59	127	25.2%
Annual mileage travelled		
< 7500	104	20.7%
7500 to 10,000	89	17.7%
10,000 to 15,000	213	42.3%
> 15,000	97	19.3%

Brackets are created for each continuous variable, except for number of at fault accidents. 431 out of 503 drivers did not have any at fault accident in the past three years. Among those that have had accidents, most had only 1 accident. 7 drivers have reported 2 at fault accidents in the past three years. Most drivers have at least 10 years of driving experience, and female drivers account for about 53 percent of our sample. Of 503 drivers, 231 are in the age group 30 to 44, representing 45.9% of drivers. 81% of the drivers drive less than 15,000 miles per year, with the majority driving between 10,000 to 15,000 miles annually.

4.4. **Empirical Analysis**

We use generalized linear models (GLMs) to examine the relationship between a variety of risk factors and the probability of auto accidents (McCullough and Nelder, 1989). GLMs are well accepted in actuarial application, and are the primary method many insurance companies use in pricing auto insurance (see, for example, Renshaw, 1994). The framework of GLMs allows for various assumptions about the distributions of the response variables. Given that insurance data is seldom normally distributed, such a technique is appropriate. We treat each trip as a distinct observation point and conduct analyses on a per-trip basis. The benefits of performing analyses at trip level include a larger sample size as well as allowing for variation on trip-level variables, our key variables of interest.

The main response variables that we use are at-fault accidents within the past three years. While in the survey we also collect data on total accidents regardless of fault, as well as accidents beyond the past three years, we feel that at-fault accidents within three years is most appropriate for several reasons. First, given that our research design relates current driving patterns with loss history, there is assumption that a person's current driving habits resemble the driving pattern displayed when past accidents were observed. A person's driving habits are likely to have changed from the pattern if they are more than three years old. Second, the use of loss history in the past three years is consistent with the common practice in the insurance industry when underwriting auto insurance policies. We also choose to use at-fault accidents, rather than total accidents, because our goal is to identify driving habits that are hazardous, and provide linkage between accidents and the driving habits which may be attributed to the accidents.

Two types of dependent variables are created for the regression analysis. The first one is a dummy variable that is coded 1 if a person has had any accident, and zero if there is no accident. The general form of our models can be expressed by the following equation:

$$\delta_{m,i}(p) = \alpha\pi_m + \lambda\gamma_{m,i} + \varepsilon_{m,j} \quad \text{(Eq. 13)}$$

For modeling binary data (e.g. accident/no accident), we use a logit link, where

$$\delta(p) = log[p/(1-p)] \quad \text{(Eq. 14)}$$

The p in the dependent variable represents the probability of having a loss, and $p/(1-p)$ represents the odds ratio. Given that a few people have reported more than 1 accident in the past, we also model number of accidents. We have tried both Negative binomial distribution as well as Poisson distribution, and report the results from Poisson regression as the dispersion is zero.

Table 8 shows the GLM results from modeling the probability and number of at-fault accidents.[6][7] We performed both Pearson and Deviance tests for our models and all tests consistently support our models and confirm that there is no evidence of lack-of-fit.

Our models include common UBI factors such as time of travel, brake, and start habits as well as contextual variables such as speeding and relative speed. We also control for traditional underwriting factors such as gender, driving experience, and vehicle mileage. Categorical variables are created for each continuous measure to generate different risk buckets. For hard brakes and hard starts, dummy variables are created identifying whether the trip has had any qualified hard brakes or hard starts. For contextual

Table 8 GLM on probability of at-fault accidents

Variables	Accident or no accident		Number of accidents	
	Coefficient (Std. error)	Relative prob. ratio	Coefficient (Std. error)	Incidence-rate ratio
Intercept	-1.688*** (0.032)	-	-1.719*** (0.029)	-
Drive more than 10 years	-0.458*** (0.029)	0.671	-0.491*** (0.025)	0.612
Female	-0.480*** (0.018)	0.658	-0.517*** (0.017)	0.597
Age between 30 and 44	-0.507*** (0.024)	0.642	-0.391*** (0.022)	0.676
Age between 45 and 59	-0.688*** (0.030)	0.545	-0.548*** (0.028)	0.578
VMT between 7,500 and 10k	0.202*** (0.033)	1.183	0.122*** (0.030)	1.130
VMT between 10k and 15k	0.134*** (0.027)	1.119	0.108*** (0.025)	1.114
VMT more than 15k	0.871*** (0.030)	1.963	0.818*** (0.026)	2.265
Peak time	0.037** (0.019)	1.032	0.014 (0.017)	1.014
Hard brake[8]	0.125*** (0.019)	1.110	0.101*** (0.017)	1.106
Hard start	0.091*** (0.019)	1.079	0.051*** (0.017)	1.052
Speeding when congestion	0.135*** (0.022)	1.119	0.121*** (0.019)	1.129
Speeding when no congestion				
Speed limit $\geq$ 60	0.276*** (0.027)	1.256	0.219*** (0.024)	1.244
Speed limit <60	0.209*** (0.021)	1.189	0.156*** (0.018)	1.169

[6] Results presented in this table use $-10ft/s^2$ and $10ft/s^2$ as the triggers for hard brake and hard start, respectively. We have also tried alternative models using $\pm$ $12ft/s^2$and $\pm$ $15ft/s^2$ and the results are similar to the ones reported here. For contextual driving factors, we have also used various thresholds, including 5mph, 8mph (reported here), 10mph over speed limit for the speeding variables. For relative speed, we used 5mph, 10mph, 30% (reported here), 50%, 70% and 100%, 1 or 2 standard of link speed. The results from alternative models can be obtained from the authors.

[7] * indicates significance at 10% level, ** indicates significance at 5% level, *** indicates significance at 1% level

[8] Results presented in this table use $-10ft/s^2$ and $10ft/s^2$ as the triggers for hard brake and hard start, respectively. We have also tried alternative models using $\pm$ $12ft/s^2$and $\pm$ $15ft/s^2$ and the results are similar to the ones reported here. For contextual driving factors, we have also used various thresholds, including 5mph, 8mph (reported here), 10mph over speed limit for the speeding variables. For relative speed, we used 10mph, 30% (reported here), 50%, 70% and 100%, 1 or 2 standard of link speed. The results from alternative models can be obtained from the authors.

Faster relative speed				
Link speed < 40mph	0.111*** (0.022)	1.098	0.083*** (0.020)	1.086
Link speed ≥ 40mph	0.046** (0.023)	1.039	0.055*** (0.020)	1.057
Slower relative speed				
Link speed < 40mph	-0.070*** (0.023)	0.942	-0.038* (0.021)	0.963
Link speed ≥ 40mph	0.387*** (0.022)	1.371	0.339*** (0.019)	1.403
Goodness of Fit Tests				
Deviance	34857.40		64455.84	
Pearson	40676.63		128858.64	

variables such as speeding and relative speed, we create dummy variables using the 75% value of each distribution as the cutoff point.[9] Specifically, we rank each of the following seven variables:

1. Percentage of a trip speeding over a specified threshold[10] during traffic congestion;
2. Percentage of a trip speeding over a specified threshold during non-congestion when speed limit is ≥ 60mph;
3. Percentage of a trip speeding over a specified threshold during non-congestion when speed limit is < 60mph;
4. Percentage of a trip faster than link speed by a specified threshold[11] when link speed < 40mph;
5. Percentage of a trip faster than link speed by a specified threshold when link speed ≥ 40mph;
6. Percentage of a trip slower than link speed by a specified threshold when link speed < 40mph;
7. Percentage of a trip slower than link speed by a specified threshold when link speed ≥ 40mph;

A variable is coded as 1 if its value is greater than or equal to the 75th percentile of the distribution, and zero otherwise. By doing so, we create seven indicator variables to be included in the regression analysis.

Each of the coefficients in our models indicates whether the particular risk factor is associated with greater accident rate. To get an intuitive explanation of how accident rate changes, we calculate the relative probability of having an accident from logit models for each variable using regression coefficients. The base probability used to standardize each probability takes the value of zero for all variables. Standardized probability allows us to see the relative probability of accidents when the corresponding variable increases by one unit.[12] For Poisson models, we calculate incident-rate ratios by exponentiation of the model coefficients. The model results have been consistent regardless of the model structure (e.g. logit or Poisson).

[9] We also tried using the mean values of the distributions as the cutoff points. The results are consistent with the ones using the 75th percentile as the cutoff points.

[10] Examples of such thresholds include 5mph, 8mph, 10mph, etc.

[11] Examples of such thresholds include 10mph, 30% of link speed, or 1 standard deviation of link speed, etc.

[12] The probability of accident from the logit model can be calculated by $p = \exp(\alpha\pi_m + \lambda\gamma_{m,j})/(1+\exp(\alpha\pi_m + \lambda\gamma_{m,j}))$. When all variables take the value of zero, the only coefficient that goes into calculation for probability is the intercept. This represents the probability of a driver having an accident who is male, younger than 30, has driven less than 10 years, drives less than 7,500 miles per year and does not possess any risky behavior in terms of trip level driving attributes. In this case, the probability of having an accident is exp(-1.688)/(1+exp(-1.688)) = 0.156. This is the base probability that we use to standardize probabilities for all other variables.

5. Numerical Analysis

5.1 Contextual driving risk analysis

Starting with traditional static underwriting factors, our results indicate that female drivers and people with more driving experience are less likely to have an accident, which is expected. Additionally, our findings show that all age groups have lower accident rates when comparing to those 29 and younger[13]. Specifically, the relative probability ratio for female is 67.1%, which suggests that the probability of accidents for a female driver is 32.9% lower than that for a similar male driver. Similarly, the probability of having an accident for a male driver in the age range 30 to 34 is 35.8% lower than the probability of accidents for a male driver who is younger than 30 years old.

With regard to vehicle mileage travelled, people who drive more than 15,000 miles per year are found to have the highest accident rate, which justifies the use of mileage for pricing auto insurance under UBI programs. Peak time travel, hard brakes and hard starts are also found to be associated with higher accident rate. Comparing to drivers in our base category, the probability of accident would be 11% and 7.9% higher if a driver tends to have hard brake and hard start, respectively. Peak time driving also increases the accident rate by 3.7%.

One of the unique contributions of our data is that we not only monitor common UBI factors, our data also include contextual risk factors. Our findings show that speeding and relative speed are both relevant to accident rate at various speed limits as well as traffic conditions. Specifically, our results show that

1. Speeding could result in an 11.9% to 25.6% greater chance of loss depending on the speed limit as well as whether or not there is congestion.
2. With regard to relative speed, we find that driving significantly faster than link speed is more likely to have accidents, regardless of link speed.
3. When link speed is 40mph or higher, our findings suggest that driving significantly slower than link speed is a significant risk multiplier and may increase the probability of accident by 37.1%. One possible explanation is that slower driving in this condition may be associated with being inattentive to traffic conditions, which would increase risk.
4. Driving slower than traffic when link speed is below 40mph, however, is found to be associated with lower accident rate. In other words, when traffic speed is slow, driving with caution at a slower speed than link speed is viewed favorably.

Combined together, our results support the contention that differentiating driving risks under various contexts in terms of traffic flow is important when determining risks.

5.2 Premium implications

The findings on how different risk factors affect the probability of auto accidents has significant implications for insurance companies. Insurance companies that do not separate high risks from low risks may suffer from adverse selection when competing with other insurers that do differentiate risks when pricing auto insurance. The following example illustrates the extent of premium undercharge when an insurer omits speeding information as an underwriting factor.

Using the regression coefficients from the logit model in Table 8, we calculate and compare the hypothetical premiums for drivers with different driving patterns in terms of the tendency to drive faster than the speed limit and present the results in Table 9. We first calculate the probability of having a loss for each risk category. The comparison is based on the assumption that the driver is 40 years old, has over 10 years of driving experience, and drives 12,000 miles per year. We assume the driver does not have qualified high risk factor in terms of braking/starting habits, time of travel and relative speed. We report

[13] Those who are 29 or younger are in the omitted category.

the relative premiums for male and female drivers separately. Assume the actuarially fair premium for a male driver that does not tend to drive significantly faster than the speed limit is $1,200. Depending on whether a driver tends to drive faster than the speed limit in different situations, the actuarially fair premium for a male driver ranges from $1,200 to $2,096. For a female driver with the same characteristics, premiums range from $764 to $1,364. Premium is the lowest when a driver never or seldom drives faster than the speed limit and premium is the highest when a driver tends to be speeding under all circumstances. Speeding when there is congestion as well as when there is no congestion both would contribute to higher premiums. The increase in premium is especially significant when one tends to be speeding during open traffic flow when speed limit is high.

Given that various speeding-related variables are found to affect the probability of accidents, an insurer that does not differentiate premiums according to these factors may incur adverse selection resulting from low risk drivers choosing to buy from other insurers. In Table 10 we show the potential underwriting loss if an insurer were to omit speeding information in underwriting. Specifically, we show the percentage of observations in each combination of speeding category in our sample, and calculate the average premium for a male driver and a female driver if an insurer were to have a group of policyholders that resembles our sample. Combining the actuarially fair premium reported in Table 8 and the percentage of observations in each risk level shown in Table 10, we calculate the weighted average premium for a male

Table 9 Premium comparison by incorporating speeding variables

Congestion	No Congestion		Male		Female	
	Speed limit ≥ 60	Speed limit <60	P(loss)[14]	Premium	P(loss)	Premium
no	no	no	7.5%	$1,200	4.7%	$764
yes	no	no	8.4%	$1,359	5.4%	$869
no	yes	no	9.6%	$1,545	6.2%	$992
no	no	yes	9.0%	$1,453	5.8%	$931
yes	yes	no	10.8%	$1,744	7.0%	$1,125
yes	no	yes	10.2%	$1,642	6.6%	$1,057
no	yes	yes	11.6%	$1,862	7.5%	$1,205
yes	yes	yes	13.0%	$2,096	8.5%	$1,364

Table 10 Premium effect due to adverse selection resulting from omitting speeding variables

Congestion	No Congestion		% of observation			
	Speed limit ≥ 60	Speed limit <60	Assumed distribution		Actual distribution	
			Male	Female	Male	Female
no	no	no	68.3%	80.5%	0.0%	0.0%
yes	no	no	6.3%	6.7%	19.9%	34.4%
no	yes	no	2.3%	1.8%	7.2%	9.4%
no	no	yes	16.2%	7.8%	51.1%	39.8%
yes	yes	no	2.2%	1.2%	6.8%	6.2%
yes	no	yes	2.7%	1.7%	8.6%	8.6%
no	yes	yes	1.9%	0.2%	5.9%	0.8%

[14] The probability of accident can be calculated by $\exp(\alpha\pi_m + \lambda\gamma_{m,j})/(1+\exp(\alpha\pi_m + \lambda\gamma_{m,j}))$ using regression coefficients from the logistic regression. For example, the probably of having an accident for a 40-year-old male driver who has over 10 years of driving experience, drives 12,000 miles per year and does not have qualified high risk factor in terms of braking/starting habits, time of travel and relative speed is exp(-1.688-0.4581-0.507+0.134)/(1+exp(-1.688-0.4581-0.507+0.134)) = 0.075. The same process can be applied to other combination of driver and driving attributes.

yes	yes	yes	0.1%	0.2%	0.5%	0.8%
Average premium:			$1,296	$799	$1,504	$944
Under-charged premium:					$208	$145

driver to be $1,296. This represents the premium an insurer who does not include speeding information in underwriting would charge each male policyholder before loading. This insurer would charge a similar female driver $799 before loading. However, the pricing would have been too low as the actual distribution of policyholders in each risk category in this insurer's pool would be different from the assumed distribution due to adverse selection. Assuming consumers stay informed, male drivers whose actuarially fair premiums are lower than $1,296 and female drivers whose actuarially fair premiums are lower than $799 would likely drop out of this insurer's pool. As a result, the premiums this insurer should have charged, based on the actual distributions of policyholders in the pool, are $1,504 for males and $944 for females. In other words, the insurer would have undercharged a male driver $208 and a female driver $145 resulting in 16% and 18.1% underwriting loss for male and female drivers, respectively.

Similar comparisons like the ones shown above can also be made to show the marginal benefits of incorporating hard brakes, hard starts, time of travel or relative speed factors. For example, Table 11 shows the potential undercharge of premiums if an insurer were to omit time of travel, hard brake or hard start variables in its auto insurance pricing. We consider four scenarios: when an insurer

1. does not include peak time travel as an underwriting factor,
2. does not include hard brake as an underwriting factor,
3. does not include hard start as an underwriting factor, or
4. does not include either hard brake or hard start as underwriting factors.

Table 11 Premium effect for males due to adverse selection resulting from omitting peak time travel, hard brake, and hard start

Peak Time	Hard Brake	Hard Start	Assumed Distribution	Actual Distribution			
				Excludes hard brake	Excludes hard start	Excludes hard brake & hard start	Excludes peak time
No	No	No	38.4%	0.0%	0.0%	0.0%	0.0%
Yes	No	No	18.2%	0.0%	0.0%	0.0%	60.6%
No	Yes	No	9.9%	36.6%	0.0%	0.0%	0.0%
No	No	Yes	11.6%	0.0%	40.0%	0.0%	0.0%
Yes	Yes	No	4.4%	16.1%	0.0%	0.0%	14.5%
Yes	No	Yes	4.6%	0.0%	15.8%	0.0%	15.3%
No	Yes	Yes	9.9%	36.6%	34.2%	77.4%	0.0%
Yes	Yes	Yes	2.9%	10.7%	10.0%	22.6%	9.7%
Average premium:			$1,285	$1,414	$1,387	$1,474	$1,306
Underwriting loss:				$130	$102	$189	$22

According to the assumed distribution, the average premium for the pool that contains 40-year-old male drivers that drive 12,000 miles per year is $1,285. When an insurer does not consider hard brake as a risk factor, and combines drivers that frequently have hard brakes with those that do not have hard brakes in the same risk category, policyholders with frequent hard brakes would be subsidized by those that do not. Adverse selection theory suggests that drivers that do not have hard brakes will not be likely to stay with this particular insurer and eventually the insurer's business will only retain drivers that frequently drive with hard brakes. The reallocation of distribution will result in 36.6% of observations in the category that has hard brakes, but does not have hard starts and does not travel during peak time, 16.1% of the observations in the category that travels frequently during peak time and also has qualified hard brakes

but does not have hard starts, 36.6% of the observations in the category that has qualified hard brakes and hard starts but does not travel during peak time, and 10.7% in the category that not only travels frequently during peak time, but also have qualified hard brakes as well as hard starts.

The same logit can be applied to the other three scenarios. As can be seen from Table 11, an insurer that does not include hard brakes as an underwriting factor should charge a premium of $1,414 as the insurer's pool will now contain only policyholders that tend to drive with hard brakes. As a result, the insurer would incur an underwriting loss of $130 per male driver. Similarly, if an insurer were to exclude hard start as an underwriting factor, the insurer may incur an underwriting loss of $102 per male driver. If an insurer were to exclude multiple underwriting factors that affect accident rate, for example, both hard brake and hard start patterns, the insurer could incur an underwriting loss of $189 per male policyholder.

Similarly, we can calculate and compare the hypothetical premiums insurers would charge based on whether they incorporate relative speed information or not. Using the distributions of drivers in each risk category depending on how they drive relative to link speed, we have calculated the average premiums to be $1,318 for males and $868 for females, as reported in Table 12. However, an insurer that pool high risks with low risks may attract those that tend to drive faster than link speed in its pool as premiums charged by this insurer would be lower than the actuarially fair premiums according to the true risk level. The actuarially fair premiums based on the distributions of drivers in each risk category after taking adverse selection effect into consideration are $1,715 for males and $1,129 for females. As a result, the insurer can expect underwriting loss of $397 and $261 for males and females, respectively, which is equivalent to 30.1% undercharge.

T*able 12 Premium effect due to adverse selection resulting from omitting relative speed variables*

Faster than link speed		Slower than link speed		% of Distribution			
Link speed < 40mph	Link speed ≥ 40mph	Link speed < 40mph	Link speed ≥ 40mph	Assumed distribution		Actual distribution	
				Male	Female	Male	Female
No	No	No	No	39.7%	28.6%	0.0%	0.0%
Yes	No	No	No	10.2%	9.0%	7.9%	0.0%
No	Yes	No	No	10.1%	16.3%	0.0%	0.0%
No	No	Yes	No	12.6%	10.2%	0.0%	0.0%
No	No	No	Yes	10.0%	10.9%	10.1%	10.9%
Yes	Yes	No	No	1.4%	0.5%	8.2%	8.8%
Yes	No	Yes	No	3.1%	1.3%	0.0%	0.0%
Yes	No	No	Yes	0.5%	0.3%	11.1%	12.1%
No	Yes	Yes	No	1.7%	5.9%	0.0%	0.0%
No	Yes	No	Yes	4.0%	8.5%	10.5%	11.4%
No	No	Yes	Yes	5.0%	5.3%	9.5%	10.2%
Yes	Yes	Yes	No	0.2%	0.0%	0.0%	0.0%
Yes	Yes	No	Yes	0.3%	0.4%	11.6%	12.6%
No	Yes	Yes	Yes	0.8%	2.4%	9.9%	10.7%
Yes	No	Yes	Yes	0.4%	0.2%	10.5%	11.4%
Yes	Yes	Yes	Yes	0.0%	0.2%	10.9%	11.8%
			Average premium:	$1,318	$868	$1,715	$1,129
			Underwriting loss:			$397	$261

6. Conclusions

The auto insurance market has changed rapidly in the recent years due to advances in insurance telematics technology. More and more insurers are interested in implementing UBI programs that take advantage of the newly available GPS data. However, lacking the funding for the technology to collect driving data as well as the unfamiliarity with the insurance telematics data have limited the accessibility of UBI. In this research we collect and examine second-by-second GPS data, and showcase how GPS data can be integrated into existing or new auto insurance pricing structures. Our findings confirm that mileage, peak time travel as well as driving behavior such as braking or starting habits are highly correlated with accident rate. We also find that *contextual driving factors* such as speeding and relative speed are important risk factors. Our findings establish the actuarial foundation for UBI pricing and would benefit insurance companies that are interested in incorporating telematics data into their auto insurance premium rating system.

There are a few limitations that may be improved upon by future studies. First, given that we did not observe any actual auto accidents from our users within the observation period, we use auto accident history as a proxy for current accidents. Working with an insurance company on a larger scale data collection that involves more drivers may allow for the observation of actual accidents. Second, premium rates are composed of loss frequency combined with loss severity. In the current study we assume constant loss severity when performing premium comparison based solely on loss frequency. Future studies that also incorporate severity dimension will enable a refined analysis on the connection between personalized driving risk and auto premium rates.

While our study has demonstrated the potential underwriting loss an insurance company may incur if it does not price auto insurance according to a driver's respective loss potential, one should also factor in the cost of incorporating detailed risk factors when evaluating the feasibility of a UBI program. Potential costs to be considered include technology, communication and personnel expenses, just to name a few. Depending on the size of the pool in each risk category, there may also be potential misclassification consideration. Nonetheless, with insurance telematics technology being available, insurance companies that are capable of offering UBI options will have the advantage of further differentiating drivers based on their true driving risks.

7. Acknowledgments

This research is supported by Federal Highway Administration (FHWA) Broad Agency Announcement "Pay-As-You-Drive-And-You-Save (PAYDAYS) Insurance Actuarial Study" project. Contract award number DTFH61-13-C-00033. We thank Allen Greenberg from FHWA for his insight and expertise that greatly assisted the research.

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